Evaluation of feature extraction methods with ensemble learning for breast cancer classification

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***Abstract*—A breast cancer is known as one of the most dangerous disease of the death causes among aged 40-55 women. We need to develop a computer aided diagnosis system for breast cancer classification. In the previous study, the random forest of the ensemble learning method was reported to be one of promising classifiers for classifying breast cancers using a Wisconsin Diagnostic Breast Cancer(WDBC) dataset. Furthermore, we found the effectiveness of the random forest with recursive feature elimination(RFE) for breast cancer classification, compared to another ensemble learning methods, XGBoost and LightGBM(LGBM). The RFE is known as a feature selection method. In our experiments, however, there was no comparison with the feature extraction of principal component analysis(PCA). This paper presents the evaluation of feature extraction methods with ensemble learning for breast cancer classification on the WDBC dataset.**

***Index Terms*—Wisconsin Diagnostic Breast Cancer(WDBC) dataset, Breast cancer classification, Random forests, XGBoost, LightGBM(LGBM), Principal component analysis(PCA), Recursive feature elimination(RFE)**

I. Introduction

A breast cancer is known as one of the most dangerous disease of the death causes among aged 40-55 women [1]. Therefore, in order to detect breast cancer early, we has desired designing a computer aided diagnosis system for breast cancer classification. In this study, we used a Wisconsin Diagnostic Breast Cancer(WDBC) Data Set [2]. This data consists of 212 malignant and 357 benign, totally 569 samples. This data has 30 features, which have 10 attributes, radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension, and three types of statistics, mean, error, and worst. We should note that "error" and "worst" mean the standard deviation and the average of top 3 sizes of each attribute in descending order, respectively. In the previous study [1], the random forest [3] which is known as an ensemble learning method was reported to be one of promising classifiers for classifying breast cancers on the WDBC dataset. According to the results [1], the random forest was the best among SVM(linear), k-NN($k = 3$), Naive Bayes, J48, random forest, and multilayer perceptron, in terms of the accuracy. Furthermore, from the previous study [4], we showed the effectiveness of the random forest with a recursive feature elimination(RFE) [5] for breast cancer classification on the WDBC dataset, compared to the state of the art ensemble learning techniques, XGBoost [6] and LightGBM(LGBM) [7]. We are expecting the improvement of the breast cancer classification performance by the feature extraction. In the pattern recognition field, there are 2 types of feature extraction methods, in general. One is feature selection. The other is feature extraction. In the previous study, we tried using a feature selection of the RFE for breast cancer classification. However, there was no comparison with the feature extraction of principal component analysis(PCA) or Karhunen-Loeve(KL) expansion [8] [9]. The PCA is well known as a powerful tool of the feature extraction method. Therefore, in this study, we try examining the PCA as the feature extraction for breast cancer classification. This paper presents the evaluation of feature extraction methods with ensemble learning for breast cancer classification on the WDBC dataset.

II. Methodology

First, we describe ensemble learning methods, random forests [3], XGBoost [6], and LGBM [7]. The random forest is one of the ensemble training approaches. The ensemble learning is based on decision trees. The random forest classifies a class based on the multiple outputs performed by each decision tree [3]. A decision tree itself causes over-training easily. Therefore, the random forest used by multiple decision trees is expected avoid over-training. Depending on the pattern classification problem, we could set the number of the decision trees of the random forest. Alternatively, there are the state of the art ensemble learning techniques of the XGBoost [6] and the LGBM [7]. Unlike a single decision tree, the random forest uses multiple decision trees as bagging. This relates to a parallel learning processing. On the other hand, the XGBoost and the LGBM use multiple decision trees as boosting. This learning serially processes. In general, the cost of training the XGBoost is relatively high. On the other hand, the learning cost of the LGBM aims to be cheaper than that of the XGBoost.

Second, we show the feature extraction methods. One possibility of the feature extraction is to reduce a classifier design cost, keeping the average error rate low. In general, there are two types of feature extraction methods: feature selection and feature extraction. The RFE algorithm which is a feature selection recursively eliminates the variables using permutation importance measure as a ranking criterion [5]. We could use

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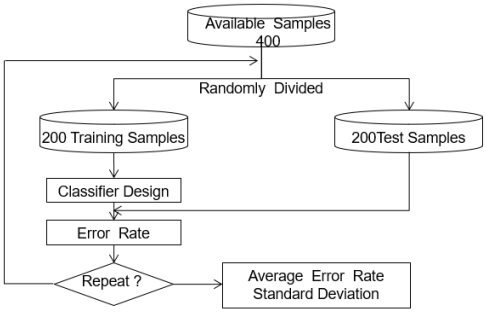

Fig. 1. Error rate estimation.

richer features selected for breast cancer classification. On the other hand, the PCA [8] [9] is a feature extraction. By doing this, we could select eigen vectors with larger eigen values which means having richer features. When it comes to before training a classifier, we transform each variable into that with keeping the mean value 0 and its standard deviation 1. Note that both the mean value and the standard deviation should be computed only by training samples not by test samples.

III. Experiments

We used 400 samples from the WDBC dataset: 200 malignant samples and 200 benign samples. In order to evaluate the classification performance of the breast cancer, we estimate the average error rate by a holdout method [8]. The holdout method maintains a statistical independency between the training and test samples. The error rate is calculated as a ratio of the number of test samples misclassified to the number of all test samples.

$$\text{Error rate} = \frac{\text{\#test samples misclassified}}{\text{\#all test samples}} \times 100(\%) \qquad (1)$$

Fig. 1 shows the error rate estimation. First, we randomly divided 400 available samples into 200 training samples and 200 test samples, respectively. Second, we trained the classifier with 200 training samples. Next, using the classifier, we computed the error rate with 200 test samples. By repeating 100 times, we obtained the average error rate and the standard deviation.

Firstly, we showed the ensemble classification performance by the random forest, XGBoost, and LGBM, in terms of the error rate [4]. Table I shows the average error rate for each classifier when the feature size is 30. The figure 30 means we used all the features provided by the WDBC dataset. We set to some parameter values from the preliminary experiments [4]. The number of the decision trees with the random forest was 100. In the XGBoost, the number of the boost rounds was 100. From the table I, we see the random forest is slightly superior to the XGBoost and LGBM classifiers [4].

Secondly, we examined the effect of ensemble learning methods of the random forest, XGBoost, and LGBM with the RFE and the PCA, respectively. We focused on the average error rate when the number of the features is 15. The figure 15 is just a half of 30 which is a full of features of the WDBC dataset. Table II shows the average error rate for each classifier with the RFE and the PCA when the feature size is 15.

TABLE I
Average error rate for each classifier when the feature size is 30.

RF	XGBoost	LGBM
4.82	5.34	4.92
1.32	1.93	1.68

Upper: average error rate(%), Lower: standard deviation

TABLE II
Average error rate for each classifier with the RFE and the PCA when the feature size is 15.

	RF	XGBoost	LGBM
RFE	4.66	5.16	4.92
	1.55	1.54	1.36
PCA	5.27	5.04	4.89
	1.50	1.54	1.36

Upper: average error rate(%), Lower: standard deviation

From the result of tables I and II, the random forest with the RFE very slightly outperforms the others. The minimum average error rate was 4.66%.

IV. Conclusion

This paper presented a positive effect of the random forest with the RFE for breast cancer classification on a WDBC dataset, compared to ensemble learning approaches of the XGBoost and the LGBM. From the results of our limited experiments for breast cancer classification, the random forest classifier with the RFE was very slightly superior to or comparable with the others. Perhaps the use of the random forests could be better if a computer was capable of parallel computing. In the future study, we will try creating richer features of using existing features for breast cancer classification. Furthermore, we would like to tackle to win the fight against cancer.

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